

COL2010: Data Science

End-Semester Examination — Module 2: Data Cleaning and Preprocessing

SET A

Time: 3 Hours

Total Marks: 100

Instructions to candidates.

- This paper contains **15** questions. Attempt **all** questions.
- All numerical answers must retain at least four significant figures unless an exact closed form is requested.
- All claims of the form “prove”, “show”, or “derive” require a fully rigorous argument; a correct final formula with no justification earns no credit.
- State any assumption you invoke that is not already given in the question.
- Calculators are permitted; symbolic computer algebra devices are not.

Q1. A bank models each customer as a heterogeneous record and constructs a numerical feature vector by encoding *Age* and *Annual Income* together with a diagonal covariance description of their joint natural variation, $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2)$ with $\sigma_1, \sigma_2 > 0$ possibly replaced by a full covariance matrix Σ .

- (a) Prove rigorously that for $\Sigma \succ 0$ (symmetric positive definite), the Mahalanobis distance

$$d_M(x, y) = \sqrt{(x - y)^\top \Sigma^{-1} (x - y)}$$

defines a genuine metric on $\mathbb{R}^d$: verify non-negativity, identity of indiscernibles, symmetry, and the triangle inequality. (*Hint:* reduce to the Euclidean case via the whitening map $z = \Sigma^{-1/2}(x - y)$ and invoke the already-established triangle inequality for $\|\cdot\|_2$.)

- (b) Suppose two of the encoded features are perfectly linearly dependent (e.g. *Annual Income in Rupees* and *Annual Income in US Dollars* both included as separate columns), so that Σ is singular. Explain precisely which step of your proof in (a) breaks down, and what practical remedy (in terms of Σ^{-1}) is required.

[8 marks]

Q2. The following table records the annual income (in Rs. '000) of $n = 9$ customers in a single city branch:

210, 225, 230, 240, 245, 250, 260, 275, 640.

Compute Q_1 , the median, Q_3 , and the interquartile range $IQR = Q_3 - Q_1$ using the median-of-halves convention (excluding the overall median when n is odd). Hence compute the lower and upper Tukey fences $Q_1 - 1.5 IQR$ and $Q_3 + 1.5 IQR$, and state, with justification, which observation(s) are flagged as outlier candidates. **[4 marks]**

Q3. Consider a numeric column of n observations in which a fraction $\varepsilon \in (0, 1)$ of the entries are *coordinated contaminants*, all recorded at the same extreme value M , while the remaining $1 - \varepsilon$ fraction take values confined to a bounded interval whose width is negligible compared to M (so that, asymptotically as $M \rightarrow \infty$, the clean values may be treated as concentrated at 0).

- (a) Derive a closed-form expression for the population mean $\mu(M)$ and standard deviation $\sigma(M)$ of the sample as a function of ε and M , in the limit described above.

- (b) Hence show that the ordinary z -score of a contaminated point,

$$z(M) = \frac{M - \mu(M)}{\sigma(M)},$$

converges to a **finite limit** as $M \rightarrow \infty$ for any fixed $\varepsilon \in (0, 1)$, and derive this limit in closed form as a function of ε alone.

- (c) Using the standard flagging rule $|z| \geq 3$, solve for the critical contamination fraction ε^* above which the classical 3σ rule can **never** flag the contaminants, no matter how extreme M becomes. State the numerical value of ε^* .
- (d) Now consider the median and the Median Absolute Deviation (MAD) computed on the same contaminated sample. Prove that, provided $\varepsilon < 0.5$, the median remains equal to a clean value and the MAD remains bounded independently of M , so that the modified z -score $z_{\text{mod}} = 0.6745 (M - \text{median})/\text{MAD}$ diverges to infinity as $M \rightarrow \infty$. Reconcile this with your answer to (c) by explaining precisely why the “breakdown point” of the mean/standard-deviation pair and that of the median/MAD pair differ so sharply, distinguishing the single-point-breakdown notion from the fixed-contamination-fraction asymptotic notion used above.

[9 marks]

- Q4.** Two customers are described by the raw feature pairs (Age in years, Annual Income in Rs. '000):

$$P = (28, 620), \quad Q = (52, 980).$$

- (a) Compute the raw Euclidean distance $d_2(P, Q)$ and identify which coordinate dominates it.
- (b) The branch-wide statistics are $\mu_{\text{age}} = 40$, $\sigma_{\text{age}} = 10$ and $\mu_{\text{income}} = 750$, $\sigma_{\text{income}} = 150$. Apply z -score standardization to both coordinates of P and Q , and recompute the Euclidean distance in standardized coordinates.
- (c) Explain, with explicit reference to your two computed distances, why a k -NN-based imputation or classification step performed *before* scaling can silently ignore the Age attribute, and why this is a synthesis failure of both the scaling and the distance-metric concepts from the syllabus.

[6 marks]

- Q5.** A marketing team encodes each customer’s declared interests as a binary vector over the tag set {Travel, Sports, Tech, Finance, Cooking}. Customer X has vector $x = (1, 0, 1, 1, 0)$ and customer Y has vector $y = (0, 0, 1, 1, 0)$.

- (a) Compute the Jaccard similarity $J(x, y)$ and the cosine similarity $\cos(x, y)$.
- (b) A third customer Z has declared no interests at all, $z = (0, 0, 0, 0, 0)$. Using the course convention that cosine similarity between two all-zero vectors is defined as 1.0 and between one all-zero and one non-zero vector is defined as 0.0, state $\cos(z, z)$ and $\cos(z, y)$, and explain briefly why Jaccard similarity requires no such special-case convention here.

[4 marks]

- Q6.** Let $Y = (Y_{\text{obs}}, Y_{\text{mis}})$ denote the complete data for a column and let R be the missingness indicator, with missingness mechanism $P(R \mid Y_{\text{obs}}, Y_{\text{mis}}; \phi)$.

- (a) State Rubin’s formal definitions of MCAR, MAR, and MNAR in terms of this conditional probability.
- (b) Prove that under MAR (together with distinctness of the parameters governing Y and ϕ), likelihood-based inference about the data-generating parameters may validly *ignore* the missingness mechanism (the “ignorability” result), by showing that the observed-data likelihood factors into a term involving only the data model and a term involving only ϕ .
- (c) Explain precisely which step of your argument in (b) fails under MNAR.
- (d) A bank observes that high income earners disclose their income less often than low earners, i.e. $P(R = 0 \mid \text{Income})$ increases in Income itself. Classify this mechanism formally, and instantiate your general argument from (c) to explain concretely why no imputation method using only the observed columns can, even in principle, remove the resulting bias.

[8 marks]

Q7. An analyst applies iterative (round-robin) predictive imputation to the following five customer records, using the columns Age, Income (Rs. '000), and Education (years):

Row	Age	Income	Education
1	25	450	16
2	34	610	18
3	41	?	14
4	29	520	17
5	?	700	20

All missing cells are initialized with the column mean computed over the observed values. The cyclic predictive order is **Income** $\rightarrow$ **Age**, using the fitted (given) relations

$$\widehat{\text{Income}} = 5 \cdot \text{Age} + 20 \cdot \text{Education} + 50, \quad \widehat{\text{Age}} = 0.05 \cdot \text{Income} - 2 \cdot \text{Education} + 40.$$

- (a) State the two initial fill values.
- (b) Carry out one full iteration (update Row 3’s Income, then Row 5’s Age), showing all substituted numbers.
- (c) Explain, with specific reference to which columns serve as predictors for each target column in this particular instance, why a second iteration leaves both imputed values unchanged here, and state a general structural condition on the predictor sets under which more than one iteration would genuinely be required to reach convergence.

[6 marks]

Q8. An entity-resolution system computes pairwise similarity scores for three candidate customer records A, B, C : $s(A, B) = 0.72$, $s(B, C) = 0.70$, $s(A, C) = 0.10$. Pairs are classified using lower threshold 0.40 and upper threshold 0.65 (below lower: nonmatch; above upper: match; between: review).

- (a) Classify each of the three pairs.
- (b) The system forms entities by taking connected components of the “match” graph. State the resulting clustering of $\{A, B, C\}$ and explain the transitivity problem this exposes.

- (c) Define the natural correlation-clustering objective $\text{Obj}(\mathcal{C}) = \sum_{i,j \text{ in same cluster of } \mathcal{C}} (s(i,j) - \theta)$ with reference threshold $\theta = 0.5$. Compute Obj for the connected-components clustering $\{A, B, C\}$ and for the alternative clustering $\{A, B\}, \{C\}$, and conclude formally that connected-components clustering is *not* optimal for this objective.
- (d) Generalize: consider a path of n records $v_1, v_2, \dots, v_n$ in which every adjacent pair scores s_0 (just above the upper threshold) and every non-adjacent pair scores a floor value f (below the lower threshold). Derive, as an exact function of n , s_0 , f , and θ , the value of Obj for the single connected-component clustering versus the best clustering that keeps only adjacent pairs together, and show that the gap between the two grows as $\Theta(n^2)$. Interpret this result as a formal statement about the danger of transitive closure over a non-transitive relation.

[10 marks]

- Q9.** A retail dataset records transaction bill amounts (in Rs.) for ten customers with a strongly right-skewed distribution: 18, 22, 27, 35, 48, 65, 95, 180, 460, 1850 (all in Rs. '000, say). Compute $\log_{10}$ of each value to two decimal places, and hence state the factor by which the raw range is compressed by the transform, quoting both the raw ratio $\max/\min$ and the transformed range $\log_{10}(\max) - \log_{10}(\min)$.

[4 marks]

- Q10.** A data-science team fits a **MinMaxScaler** and a **SimpleImputer**(median strategy) on the *entire* dataset (training rows *and* test rows combined) before splitting, rather than fitting on the training rows only.

- (a) Let $n_{\text{tr}}, n_{\text{te}}$ be the training and test set sizes and let $\mu_{\text{tr}}, \mu_{\text{te}}$ be the true population means of some numerical column restricted to the train and test subpopulations respectively (which may differ, e.g. under a temporal train/test split). Derive an exact expression for the bias $\mathbb{E}[\hat{\mu}_{\text{full}}] - \mu_{\text{tr}}$ of the leaked full-data mean estimator relative to the correct train-only estimator, in terms of $n_{\text{tr}}, n_{\text{te}}, \mu_{\text{tr}}, \mu_{\text{te}}$, and show it is zero if and only if $\mu_{\text{tr}} = \mu_{\text{te}}$.
- (b) Explain formally why this non-zero bias term implies that any test-set error metric computed downstream is an *optimistically biased* estimator of true generalization error whenever the imputed/scaled feature is used by the model, even though no test *label* was ever touched.
- (c) A categorical column *City* contains a value at test time that never appeared in the training rows. Give the precise, leak-free specification of how a one-hot encoder fitted only on training categories must represent this row, and explain why fitting the encoder on the union of train and test categories — even though it touches no numerical statistic — is still a form of data leakage.

[8 marks]

- Q11.** Two candidate duplicate customer records are compared on four fields. The comparison vector is: name similarity = 0.95 (Jaro–Winkler), phone agreement = NaN (one record's phone is missing), city agreement = 1 (agree), date-of-birth agreement = 0 (disagree). The supplied field weights are name 0.4, phone 0.3, city 0.2, dob 0.1.

- (a) Compute the weighted-mean comparison score using *only* the available (non-missing) fields, with weights renormalized over those fields.
- (b) Classify the pair against thresholds lower = 0.5, upper = 0.75.
- (c) Explain, with reference to your computation, why treating the missing phone comparison as 0 (disagreement) rather than NaN would be methodologically incorrect.

[6 marks]

Q12. A heart-rate-style feature column for $n = 7$ customers (measuring, say, session duration in an app, in minutes) is: 12, 13, 11, 14, 12, 13, 90.

- (a) Compute the mean, standard deviation, and the ordinary z -score of the value 90.
- (b) Compute the median, the MAD, and the modified z -score of the value 90.
- (c) Using thresholds $|z| \geq 3$ and $|z_{\text{mod}}| > 3.5$, state which method(s) flag 90 as an outlier, and relate the disagreement explicitly to the masking phenomenon you analyzed in Q3.

[4 marks]

Q13. For $x, y \in \mathbb{R}^d$, the Minkowski distance of order $p \geq 1$ is $d_p(x, y) = \left(\sum_{j=1}^d |x_j - y_j|^p \right)^{1/p}$. Let $\Delta = \max_j |x_j - y_j|$.

- (a) Prove the two-sided bound $\Delta \leq d_p(x, y) \leq d^{1/p} \Delta$ for every finite $p \geq 1$, where d is the dimension.
- (b) Hence give a rigorous limiting argument (a squeeze argument, not merely an appeal to intuition) showing that $\lim_{p \rightarrow \infty} d_p(x, y) = \Delta = d_\infty(x, y)$.
- (c) Explain why the factor $d^{1/p}$ in your bound shows that the *rate* at which d_p approaches d_∞ depends on the dimension d , and briefly discuss the practical implication for choosing p in high-dimensional outlier scoring.

[8 marks]

Q14. Two blocking passes are used to generate duplicate-candidate pairs among $n = 6000$ standardized customer records. Pass 1 (same canonical city and birth year) produces blocks of sizes 50, 80, 45, 25; all other records are singleton blocks. Pass 2 (same last-four phone digits) produces blocks of sizes 12, 9, 15, 7, 20. The two candidate-pair sets overlap in exactly 90 pairs.

- (a) Compute the number of candidate pairs generated by each pass, and the size of their union.
- (b) Compute the reduction ratio $RR = 1 - \frac{\# \text{ candidate pairs}}{\binom{n}{2}}$ for the union.
- (c) Suppose the ground truth contains 500 true duplicate pairs, of which Pass 1 alone captures 430, Pass 2 alone captures 300, and both passes jointly capture 260. Verify, via inclusion-exclusion, the pair completeness PC of the union, and state its numerical value.
- (d) Prove the general probabilistic claim: if a true duplicate pair is captured by Pass i independently with probability p_i , then the probability it is captured by the union of the two passes, $1 - (1 - p_1)(1 - p_2)$, is at least $\max(p_1, p_2)$, and use this to argue why combining two “roughly independent” blocking keys is a favourable trade against the comparatively small additional candidate-pair cost.

[6 marks]

Q15. A branch manager gives you the following eight (already standardized) customer records:

ID	Age	Height (cm)	Weight (kg)	Income (Rs. '000)	Name (standardized)
1	34	170	68	620	amit sharma
2	29	165	60	540	amit sharma
3	45	168	65	?	sunita rao
4	51	172	70	1450	vikram sen
5	38	180	42	610	neha gupta
6	60	166	63	?	d. mehta
7	41	171	66	590	sunita rao
8	33	169	64	605	priya nair

You are additionally given: among the six rows with observed Income, the mean Age is 37.67; among the two rows with missing Income, the mean Age is 52.5. From the other seven rows (excluding Row 5), $\mu = (\mu_H, \mu_W) = (169.4, 65.4)$ and

$$\Sigma = \begin{bmatrix} 16.8 & 9.5 \\ 9.5 & 7.2 \end{bmatrix}.$$

Rows 3 and 7 are suspected to be the same real customer, with comparison vector: name similarity 0.95, phone agreement NaN, city agreement 1, date-of-birth agreement 0, using the weights and thresholds of Q11.

- Using the age-conditioned evidence given, state what can and cannot be formally concluded about the missingness mechanism of Income, being careful to note the standard caveat about ruling out MNAR from observed data alone.
- Compute, by hand, the Mahalanobis distance of Row 5's (Height, Weight) = (180, 42) from μ using Σ^{-1} , showing the inverse-covariance computation explicitly, and comment on whether Row 5 would be flagged as a multivariate outlier.
- Using the entity-resolution machinery of Q11, decide whether Rows 3 and 7 should be merged, and construct the resulting golden record's Income field using the course's stated tie-breaking rule for numerical fields.
- Now argue formally: should entity resolution (deduplication) be performed *before* or *after* missing-value imputation in this specific case? Prove that your chosen ordering strictly dominates the alternative for at least one field in this dataset, and state the general audit-trail invariant (missingness indicator / provenance) that must be preserved regardless of the ordering chosen.

[9 marks]